

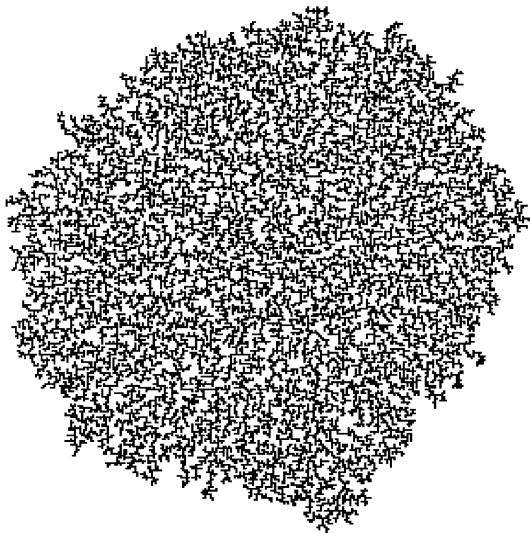
Ising Disks: Topology Preserving Glauber Dynamics

Efe Onaran¹

(Joint work w. Yuliy Baryshnikov²)

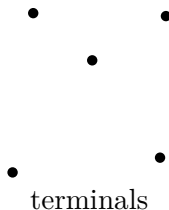
¹University of Pennsylvania

²University of Illinois Urbana-Champaign



- Yuliy Baryshnikov and Efe Onaran (2026). “Ising disks: topology preserving glauber dynamics”. *J. Appl. Comput. Topol.*

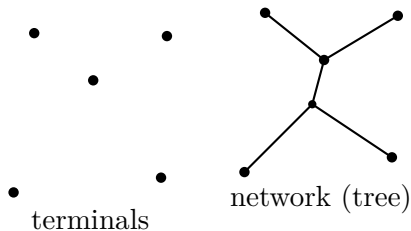
Motivation: Steiner Tree Problem



- ▶ Given fixed terminals, find the shortest connected network spanning them.
- ▶ Extra junctions are allowed; cycles are wasteful.

Gilbert and Pollak 1968

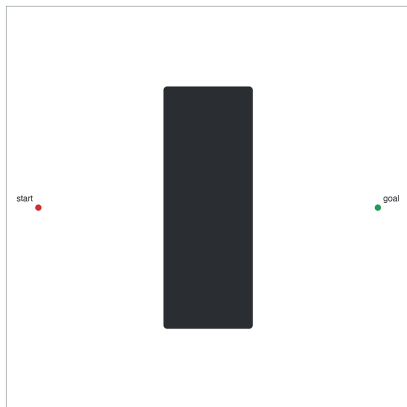
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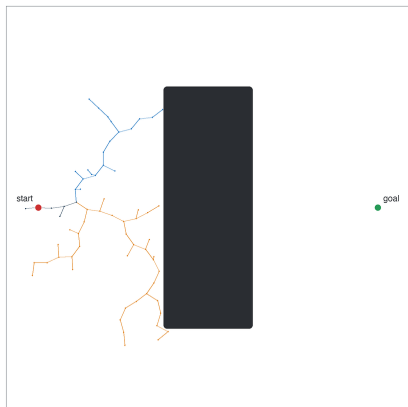
Motivation: Rapidly Exploring Random Trees



- ▶ Given start and goal states, find a collision-free path through obstacles.
- ▶ Grow one rooted tree by sampling the space and extending from the nearest existing vertex.

Simulation generated with the RRT algorithm LaValle 1998.

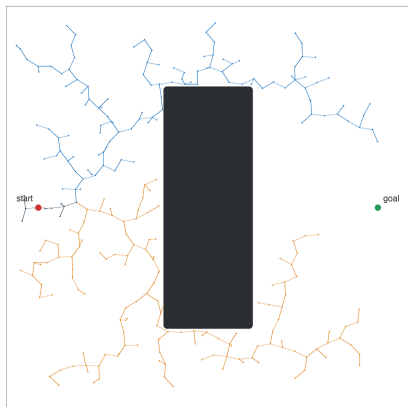
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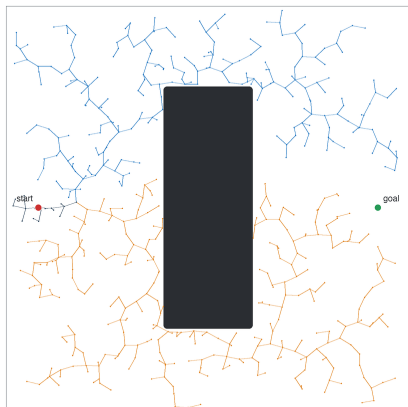
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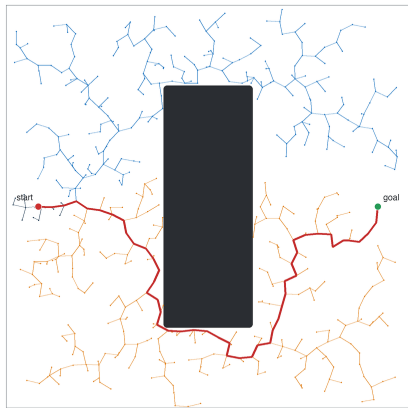
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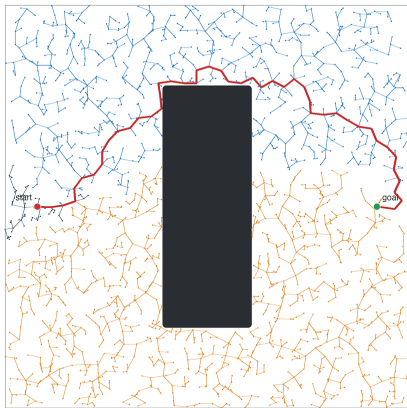
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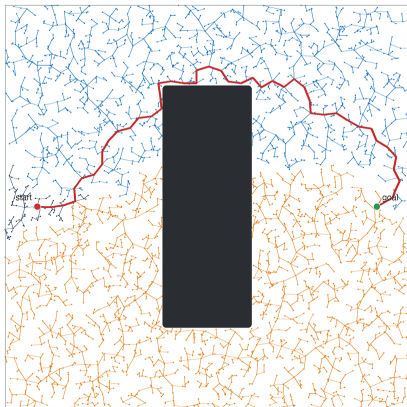
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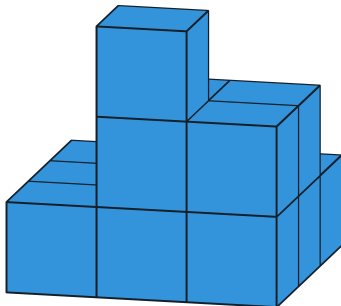
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Cubical Sets



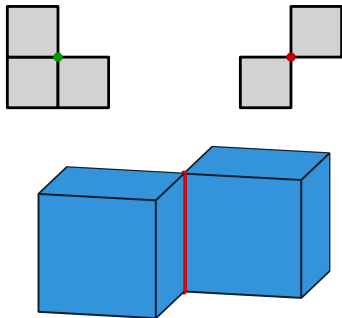
- $\mathbf{X} = \{X_1, \dots, X_m\}$ is a cubical set (complex), where

$$|X_i| = \left[l_1 - \frac{1}{2}, l_1 + \frac{1}{2}\right] \times \dots \\ \times \left[l_d - \frac{1}{2}, l_d + \frac{1}{2}\right]$$

$$(l_1, \dots, l_d) \in \mathbb{Z}^d.$$

- $|\mathbf{X}| := \bigcup_{i=1}^m |X_i|$ denotes the *geometric realization* of $\mathbf{X}$.

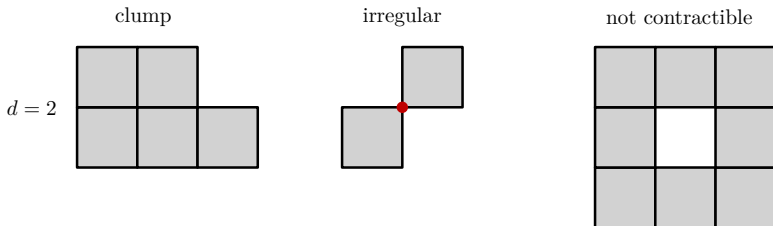
Regular Points



- ▶ Given a subset A of $\mathbb{R}^d$, a point $x \in A$ is called a regular point of A , if and only if
 - ▶ It is an interior point of A , or
 - ▶ The set $B_\varepsilon(x) \cap A \setminus \{x\}$ is contractible for all $\varepsilon > 0$ small enough.

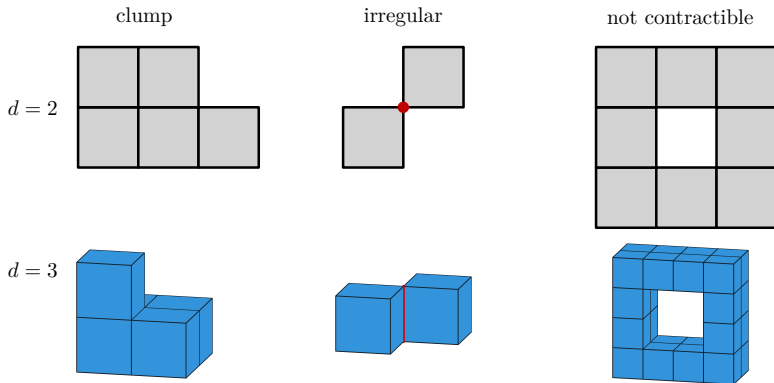
Clumps

A cubical set $\mathbf{X}$ is called a *clump* if it is regular and its geometric realization $|\mathbf{X}|$ is contractible.

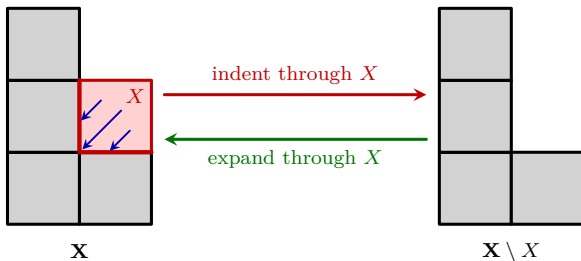


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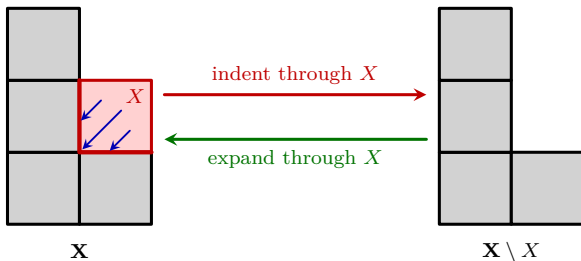
Indent/Expand actions



The clump $\mathbf{X}$ is called *indentable* through $X \in \mathbf{X}$ if

- ▶ $\mathbf{X} \setminus X$ is a clump and
- ▶ $\exists$ a strong deformation retraction $f : |\mathbf{X}| \times [0, 1] \rightarrow |\mathbf{X}|$ from $|\mathbf{X}|$ to $|\mathbf{X} \setminus X|$ satisfying $f(x, t) \in |X|$ for all $x \in |X|$ and $t \in [0, 1)$.

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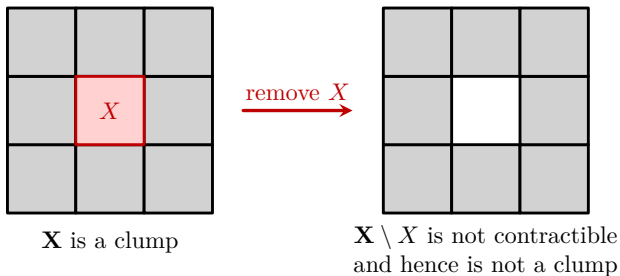
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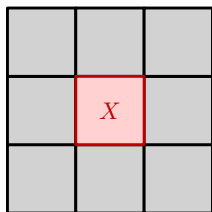
The clump $\mathbf{X}$ is called *expandable* through $Y \notin \mathbf{X}$ if

- ▶ $\mathbf{X} \cup Y$ is a clump indentable through Y .

Disallowed Indent Actions

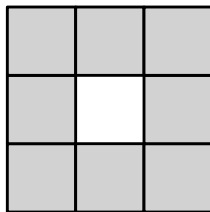


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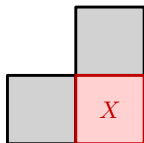


$\mathbf{X}$ is a clump

remove X →

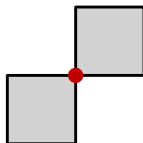


$\mathbf{X} \setminus X$ is not contractible
and hence is not a clump



$\mathbf{X}$ is a clump

remove X →



$\mathbf{X} \setminus X$ has an irregular point
and hence is not a clump

Lemma 1.

Whether a clump is indentable or not through one of its cubes is “locally decidable”.

Theorem for the planar case

Theorem 2.

In $d = 2$, every clump has at least two cubes through which it is indentable.

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Proof Idea.

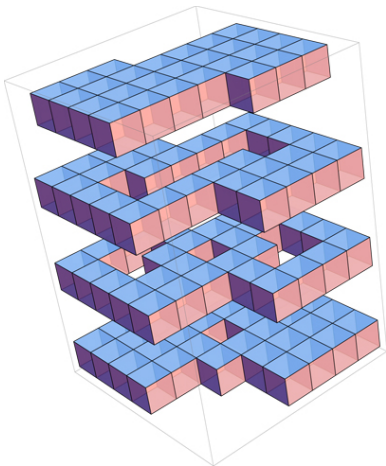
- ▶ We can roughly categorize the clumps as the “branchy” ones and “round” ones.
- ▶ For round ones, find two cubes on the border whose removal neither disconnects the clump nor creates a hole.
- ▶ For branchy ones, the longest path looks like a “line,” and its first and last cubes are indentable.



Theorem does not extend to $d \geq 3$.

Example. *Bing's house
(house with two rooms).*

It is a clump, but is not
indentable.



Shermer's cubical complex, dissected view.

O'Rourke (2011); Shermer (1988).

A Markov Process on Clumps

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- ▶ When a clock ticks at time t for a point in the cube X ,
 - ▶ If $X \in \mathbf{X}_t$, $X \neq \left[-\frac{1}{2}, \frac{1}{2} \right]^d$, and $\mathbf{X}_t$ is indentable through X :

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- ▶ If $X \notin \mathbf{X}_t$ and $\mathbf{X}_t$ is expandable through X :

$$\mathbf{X}_{t+} = \mathbf{X}_t \cup X \text{ with probability } p_{\uparrow}$$

$$d = 2$$

Theorem 3.

For $d = 2$, the Markov process is irreducible. Furthermore, it is ergodic if $\beta := \frac{p_{\downarrow}}{p_{\uparrow}} > \kappa \approx 3.9709$ with stationary distribution

$$\Pi(\mathbf{X}) \propto \begin{cases} \beta^{-\text{card}(\mathbf{X})} & \text{if } \mathbf{X} \text{ is a clump,} \\ 0 & \text{otherwise.} \end{cases}$$

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- ▶ β : “inverse-fugacity”
- ▶ κ is the “connectivity constant” of the “self-avoiding polygons”.
- ▶ Note every planar clump defines a self-avoiding polygon through its boundary.
- ▶ Let a_n be the number of self-avoiding polygons of area- n . Due to “sub-additivity”,

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \kappa.$$

$$\beta > \kappa$$



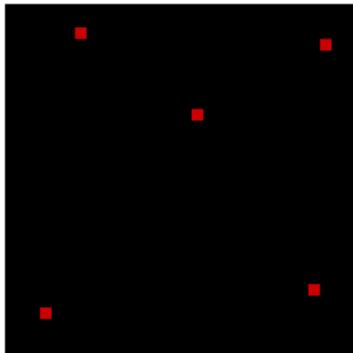
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Steiner Tree sampling



Rare events in $\beta > \kappa$ case

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Theorem 4.

If $\beta > \kappa$ then

$$N_T \xrightarrow[T \rightarrow \infty]{\text{d}} N^*.$$

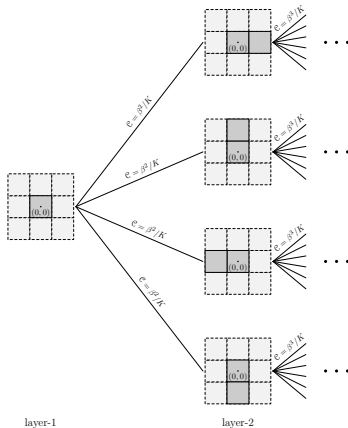
N^ : the homogeneous Poisson point process on the nonnegative real line with unit intensity.*

How does η_T scale?

Lemma 5.

$$\liminf_{T \rightarrow \infty} (\eta_T)^{1/T} \geq \frac{\beta}{\kappa}.$$

electrical network representation



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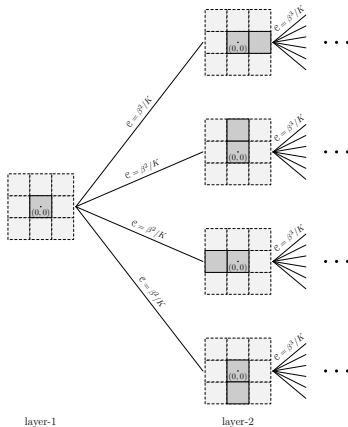
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Conjecture 6.

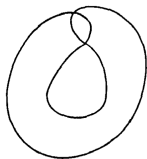
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electrical network representation



Future Work

- ▶ For $d = 2$, what if we relax the clump definition to “local” homeomorphism?
- ▶ Injectivity is not required.
- ▶ Boundary now defines a “distorted disk”.



(a)



(b)



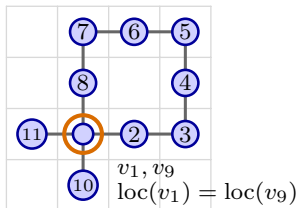
(c)

Shor and Van Wyk, *Detecting and Decomposing Self-Overlapping Curves*, Figure 1.

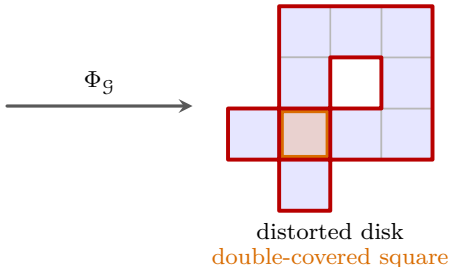
Initial Progress

- ▶ How to “discretize”?
- ▶ *Self-overlapping* polygons?

lattice net $\mathcal{G}$

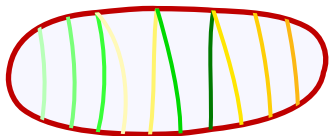


geometric realization

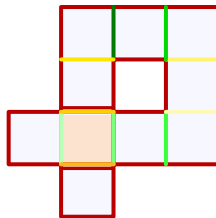


Initial Progress

grid line representation



geometric realization



distorted disk
double-covered square

Questions

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Does $\Phi_{\mathcal{G}}$ define a bijection between lattice nets and distorted (discrete) disks?

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Question 3.

Does the connectivity constant change?